Smart Multi-Sensor Wearable for Real-Time Elderly Health Anomaly and Fall Detection

Edge-Based Heartbeat Arrhythmia Classification from Wearable ECG Using Random Forest with Posture-Aware Sensor Fusion

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DECLARATION

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ABSTRACT

Heart rhythm disorders are often missed because they come and go. A short ECG taken at a clinic visit can look normal even when the patient has frequent ectopic beats at home, and long-term recorders such as Holter monitors are costly and do not raise alarms in real time. This thesis describes the arrhythmia screening component of SmartGlove, a wearable IoT system developed by Group R26-IT-108 for continuous home health monitoring. The component takes a single-lead ECG from an AD8232 sensor sampled at 250 Hz, extracts sixteen RR-interval, amplitude, interval and morphology features for the latest complete beat, and classifies it into the five AAMI classes (N, SVEB, VEB, F and Q) with a Random Forest running on a local hub without internet access. Heart-rate rules for sustained bradycardia and tachycardia are cross-checked against the posture reported by a chest belt, so that a fast pulse during running is not reported as an emergency, and the result is passed to a Master Risk Score shared with the other screening models. On 10,000 beats from 20 held-out records of a public synthetic dataset, the deployed model reached 80.7% accuracy, 68.9% balanced accuracy and 72.6% macro F1, with 94.0% recall for normal beats and 73.3% for ventricular ectopic beats, and classified one window in 3.4 ms. A comparison with six other classifiers showed that gradient boosting and a small multilayer perceptron were more accurate and far smaller. Simulation tests also revealed that the live R-peak detector double-counts beats below about 70 BPM; a band-pass detector with an adaptive threshold removed this error in every simulated case. The study shows that beat-level arrhythmia screening is practical on low-cost edge hardware, and identifies what must still be done, chiefly validation on real patient ECG, before such a tool can be trusted clinically.

Keywords: Arrhythmia detection, Wearable ECG, Random Forest, Edge computing, Sensor fusion

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Finally, I thank my family and friends for their patience and encouragement during a long and demanding final year.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AAMI	Association for the Advancement of Medical Instrumentation
ADC	Analogue-to-Digital Converter
AF	Atrial Fibrillation
BOM	Bill of Materials
BPM	Beats Per Minute
CNN	Convolutional Neural Network
CPU	Central Processing Unit
ECG	Electrocardiogram
F	Fusion beat
FR	Functional Requirement
GPIO	General Purpose Input/Output
GPU	Graphics Processing Unit
HR	Heart Rate
HTTP	Hypertext Transfer Protocol
IEC	International Electrotechnical Commission
IMDRF	International Medical Device Regulators Forum
IoT	Internet of Things
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
LO	Leads-Off
MLP	Multilayer Perceptron
N	Normal beat
NFR	Non-Functional Requirement
NMRA	National Medicines Regulatory Authority
PDPA	Personal Data Protection Act
Q	Unclassifiable beat
RF	Random Forest
RQ	Research Question
RR	Interval between two successive R peaks
SaMD	Software as a Medical Device
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SpO2	Peripheral Oxygen Saturation
SSE	Server-Sent Events

Abbreviation	Description
SVEB	Supraventricular Ectopic Beat
TC	Test Case
VEB	Ventricular Ectopic Beat
WHO	World Health Organization

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CHAPTER 1: INTRODUCTION

1.1 Background and Literature Survey

Cardiovascular disease remains the leading cause of death worldwide. The World Health Organization estimates that it caused about 17.9 million deaths in 2019, roughly 32% of all deaths, and that most of these occurred in low- and middle-income countries [1]. Sri Lanka follows the same pattern: noncommunicable diseases account for the large majority of deaths in the country, and cardiovascular disease is the largest single group among them [2]. A considerable share of cardiac deaths is sudden and is linked to disturbances of the heart rhythm [3].

An arrhythmia is any departure from the normal, regular rhythm generated by the sinus node. Some arrhythmias are harmless and occasional; others, such as frequent ventricular ectopic beats or sustained tachycardia, can signal structural heart disease, electrolyte imbalance or a risk of more dangerous rhythms. The difficulty is that many of them are paroxysmal. They appear for a few seconds or minutes and then disappear, so a 10-second ECG taken during a clinic visit can easily miss them [4]. This is the basic motivation for continuous, wearable monitoring, and it is the problem this component addresses within the SmartGlove system.

1.1.1 The electrocardiogram and heartbeat classes

The electrocardiogram (ECG) records the electrical activity of the heart from electrodes on the skin. Each normal beat produces a recognisable sequence of waves (Figure 1.1): the P wave, caused by atrial depolarisation; the QRS complex, caused by ventricular depolarisation; and the T wave, caused by ventricular repolarisation [4]. The time between two successive R peaks, the RR interval, gives the instantaneous heart rate, while the shape and timing of the waves carry information about where each beat started.

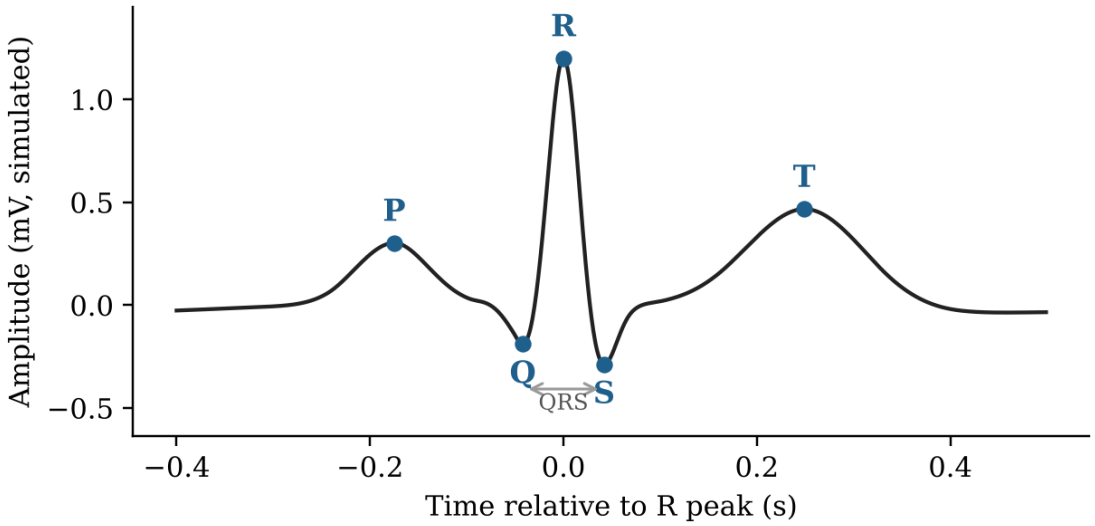


Figure 1.1: One simulated ECG beat showing the P, Q, R, S and T points used as features in this study

Source: Author, generated with NeuroKit2 [5].

To make results comparable between studies, the Association for the Advancement of Medical Instrumentation (AAMI) groups heartbeats into five classes [6]: normal and bundle branch block beats (N), supraventricular ectopic beats (SVEB), ventricular ectopic beats (VEB), fusion of ventricular and normal beats (F), and paced or unclassifiable beats (Q). An SVEB starts above the ventricles, usually in the atria, and arrives earlier than expected while keeping a fairly normal QRS shape. A VEB starts in the ventricles and typically has a wide, abnormal QRS with no preceding P wave. This difference matters for classification: VEBs differ from normal beats mainly in shape, while SVEBs differ mainly in timing. The same five-class scheme is used in this thesis.

1.1.2 Automatic heartbeat classification

Research on automatic heartbeat classification has relied for decades on the MIT-BIH Arrhythmia Database, which contains 48 half-hour two-lead recordings annotated beat by beat by cardiologists [7]. A widely cited early study by de Chazal et al. extracted RR-interval and morphology features from each beat and trained linear discriminant classifiers [8]. Its main contribution was methodological: it introduced the inter-patient evaluation scheme, in which the records used for training and testing come from

different people. Under this scheme the reported sensitivity was 77.7% for VEB and 75.9% for SVEB, with a positive predictivity of only 38.5% for SVEB, which is much lower than the near-perfect figures obtained when beats from the same patient appear in both sets [8], [9].

Later feature-based work improved on these figures through better feature selection [10], [11], [12], morphological and dynamic descriptors such as wavelet coefficients [13], and ensembles of classifiers [14]. A recurring finding in these studies, summarised in the survey by Luz et al. [9], is that RR-interval features carry most of the information needed to detect SVEBs, while QRS morphology features are more useful for VEBs. Huang et al., for example, reached an SVEB sensitivity of 91.1% under the inter-patient scheme by combining RR intervals with random projections of the beat waveform in a hierarchical classifier [15]. Even so, the positive predictivity for SVEB remained low (42.2%), which shows how hard this class is to separate from normal beats.

Deep learning changed the field from about 2016. Kiranyaz et al. trained a patient-specific one-dimensional convolutional neural network (CNN) directly on raw beats [16], and Acharya et al. classified heartbeats with a nine-layer CNN [17]. The best-known result is that of Hannun et al., who trained a 34-layer network on 91,232 single-lead ECG records from 53,549 patients and reported an average F1 score of 0.837 across twelve rhythm classes, slightly above the average of practising cardiologists (0.780) [18]. Ribeiro et al. reported similar success for 12-lead ECGs [19]. These models are accurate, but they depend on very large labelled datasets and on hardware that can run deep networks efficiently.

For tabular inputs such as a vector of beat features, the picture is different. Grinsztajn et al. compared tree-based models with deep networks on 45 tabular datasets and found that tree ensembles such as Random Forest [20] and gradient-boosted trees [21], [22] still performed as well or better on medium-sized data, while being easier to train [23]. This supports the use of tree-based models when the input is a small set of engineered features, which is the case in this study.

1.1.3 Wearable and edge ECG monitoring

Several commercial products now record the ECG outside the hospital. Patch monitors such as the Zio patch record a single lead for up to two weeks and detect more arrhythmias than a 24-hour Holter recording [24]. In the mSToPS trial, a wearable ECG patch led to new atrial fibrillation diagnoses about four times more often than usual care over four months [25]. Smartwatches have pushed this further. The Apple Heart Study enrolled 419,297 participants; 0.52% received an irregular-pulse notification, and the positive predictive value of the notification against a subsequent ECG patch was 0.84 [26]. Bumgarner et al. showed that a watch band with a single-lead ECG sensor could detect atrial fibrillation automatically [27].

These systems share two limitations for the setting of this project. First, patches and Holter monitors analyse the recording after it is returned, so they do not raise an alert while the event is happening. Second, smartwatch ECG is recorded on demand for 30 seconds when the user touches the crown, and the analysis focuses on atrial fibrillation rather than on beat-by-beat classification. In addition, most of these products send data to a cloud service for analysis, which needs a reliable internet connection and raises privacy concerns.

Edge computing moves processing close to where the data are produced, which reduces latency, keeps working when the network fails and avoids sending raw health data to remote servers [28]. In ECG telemetry, Satija et al. showed that checking signal quality on the device before transmission can reduce false alarms and wasted bandwidth in IoT health systems [29]. Low-cost front-end chips such as the Analog Devices AD8232 [30] and Wi-Fi microcontrollers such as the ESP32 [31] make it possible to build such systems at a fraction of the cost of medical-grade recorders.

1.1.4 Context-aware monitoring and sensor fusion

A wearable monitor has to cope with the fact that a high heart rate is not always a problem. A heart rate of 150 BPM is expected during running but is abnormal in a person lying in bed. Simple fixed thresholds therefore produce many false alarms in ambulatory use, which in turn leads users and carers to ignore the device. Clinical guidelines themselves define sinus bradycardia as a rate below 50 BPM and

tachycardia as a rate above 100 BPM at rest [32], [33], and the phrase "at rest" is important. Adding information about the user's activity, for example from an accelerometer, is a natural way to decide whether a fast pulse is explained by what the person is doing. In SmartGlove, a chest belt with a motion sensor reports posture, and this information is used to interpret the heart rate measured at the wrist.

1.2 Research Gap

The literature reviewed above leaves a gap between accurate offline classifiers and practical, low-cost monitors that act in real time. Table 1.1 compares representative systems on the properties that matter for SmartGlove.

Table 1.1: Comparison of representative arrhythmia monitoring approaches

Approach	Beat-level classes	Real-time alert	Works offline	Activity context	Approx. cost
Holter / patch (e.g., Zio) [24]	Yes (after recording)	No	Yes (records only)	No	High
Smartwatch ECG [26], [27]	No (mainly AF)	Partly (on demand)	Partly	No	High
Deep CNN classifiers [18], [16]	Yes	Not as a system	Needs strong hardware	No	Research
Feature-based classifiers [8], [14]	Yes	Not as a system	Yes	No	Research
This component (SmartGlove)	Yes (5 AAMI classes)	Yes	Yes (local hub)	Yes (posture)	Low

Source: Author, compiled from the cited works.

Four specific gaps can be identified.

1. **Real-time beat classification is rarely part of a complete low-cost system.** Most heartbeat classification studies stop at an offline accuracy figure on MIT-BIH [9]. They do not show how the classifier receives a live stream, how it behaves when the electrodes are loose, or how its output is turned into an alert.
2. **Rate alarms ignore context.** Fixed bradycardia and tachycardia thresholds are simple, but without knowledge of the user's posture or activity they cannot tell exercise from a genuine problem.

3. **Cloud dependence.** Many consumer and research systems need a network connection for analysis. In homes or care facilities with unreliable internet, this delays or prevents alerts and exposes raw ECG data to third parties.
4. **Results on curated data are rarely checked against the live signal path.** Classifiers are trained on features computed by one pipeline (often from 360 Hz, two-lead clinical recordings) and deployed with a different pipeline on a different sensor. The effect of this mismatch is seldom measured.

1.3 Research Problem

The research problem addressed by this component can be stated as follows:

How can heartbeat arrhythmias be classified in near real time from a single-lead wearable ECG on a low-cost local hub, without cloud processing, and how can the result be combined with heart-rate rules and posture information so that genuine abnormalities raise an alert while activity-related changes in heart rate do not?

Three sub-questions follow from this problem.

- RQ1: How accurately can a tree-based classifier assign beats to the five AAMI classes using a small set of RR-interval, amplitude, interval and morphology features, and how does it compare with other lightweight classifiers?
- RQ2: Can the classifier and its feature extraction run fast enough on CPU-only edge hardware to keep up with live telemetry?
- RQ3: How reliable is the live feature extraction pipeline across the heart rates and noise levels expected in wearable use, and what must change before the system is used on real patients?

1.4 Research Objectives

1.4.1 Main objective

The main objective of this research is to design, implement and evaluate an edge-based heartbeat arrhythmia screening component for the SmartGlove wearable system, which classifies single-lead ECG beats into the five AAMI classes, applies posture-

aware heart-rate rules, and contributes its result to the system's Master Risk Score, all without relying on cloud processing.

1.4.2 Specific objectives

1. To analyse the available ECG beat feature dataset, including its class balance, feature structure and limitations, and to design a patient-wise training and testing procedure that avoids data leakage.
2. To implement a live feature extraction routine that detects R peaks in the 250 Hz AD8232 signal and computes the sixteen beat features used by the classifier.
3. To train a Random Forest classifier for the five AAMI classes with class balancing and missing-value handling, and to compare it with alternative classifiers on accuracy, balanced accuracy, macro F1, latency and model size.
4. To implement sustained bradycardia and tachycardia rules and a posture-aware fusion step that suppresses heart-rate alerts explained by physical activity.
5. To integrate the classifier output into the SmartGlove Master Risk Score and alert pipeline.
6. To evaluate the component through held-out classification metrics, timing measurements, simulation tests of the feature extraction, and unit and scenario tests of the hub logic.

1.5 Scope and Limitations

In scope. Beat-level classification of single-lead ECG into the five AAMI classes; live feature extraction on the hub; the bradycardia and tachycardia rules; posture-aware suppression of heart-rate alerts; the hand-over of the arrhythmia result to the Master Risk Score; and the evaluation described in Chapter 3.

Out of scope. The design of the wearable hardware, the other screening models (hypoxemia, sepsis, Parkinson's tremor and stress), the correlation engine, the dashboard and the mobile application, which were developed by other members of the group. Rhythm-level diagnosis such as atrial fibrillation detection over long episodes is also outside the scope, as is any form of clinical certification.

Limitations. The classifier was trained and tested on a public dataset that turned out to be synthetic (Section 2.3). No real patient ECG with expert labels was available to the group, so the reported accuracy describes performance on that dataset and cannot be taken as clinical accuracy. Live tests were limited to simulated ECG signals and functional tests of the hub logic. SmartGlove is a screening aid and is not intended to replace medical assessment.

CHAPTER 2: METHODOLOGY

2.1 Research Approach

This research follows an applied, design-science approach: an artefact (the arrhythmia screening component) is built to solve a practical problem and is then evaluated quantitatively. The work was carried out in five phases, which overlapped in practice because problems found in later phases sent us back to earlier ones.

1. **Requirements and architecture.** The role of the component inside SmartGlove was agreed with the group, together with the data each device would send to the hub and the format of the screening output.
2. **Data analysis and model development.** The public ECG feature dataset was examined in detail, a patient-wise split was designed, and the Random Forest pipeline was trained and tuned.
3. **Live signal processing.** A feature extraction routine was written so that the hub could compute the same sixteen features from the live AD8232 signal.
4. **Hub logic and integration.** Heart-rate rules, posture-aware fusion and the hand-over to the Master Risk Score were implemented in the Flask backend and connected to the other components.
5. **Evaluation.** The classifier was evaluated on held-out records and compared with alternative models, the feature extraction was tested on simulated ECG, and the hub logic was tested with unit and scenario tests.

2.2 System Architecture

2.2.1 SmartGlove overview

SmartGlove is a two-node wearable system with a local processing hub, shown in Figure 2.1. The Smart Glove carries a MAX30102 optical sensor for heart rate and blood oxygen, an MLX90614 infrared temperature sensor and an MPU6050 motion sensor for tremor and gesture detection. The Smart Belt, worn around the chest, carries the AD8232 ECG front end, a second MPU6050 for posture and fall detection, and a

physical SOS button.Both devices are built around ESP32 microcontrollers and send their readings to the hub as JSON packets over HTTP POST on the local Wi-Fi network.

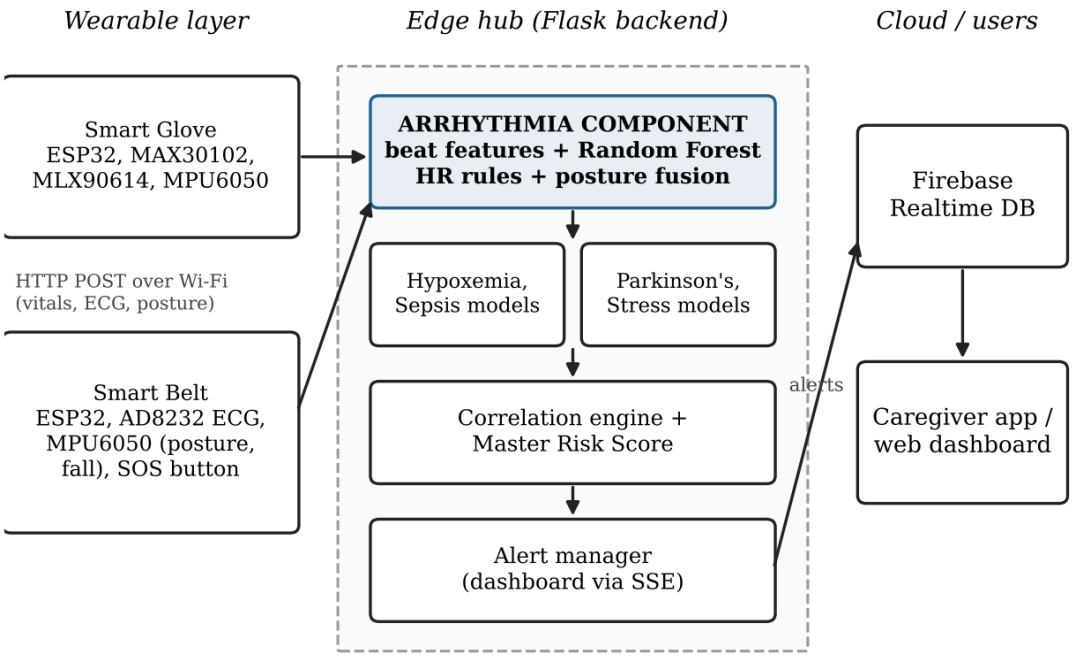


Figure 2.1: High-level architecture of SmartGlove, with the arrhythmia component highlighted

Source: Author.

The hub is a local PC or Raspberry Pi running the Python Flask backend. It hosts five screening models (arrhythmia, hypoxemia, sepsis, Parkinson's tremor and stress), several rule-based engines, a correlation engine and the Master Risk Score. Processed results are shown on a local dashboard through Server-Sent Events, and alerts are synchronised to a Firebase Realtime Database so that carers can follow them on a mobile application.

2.2.2 ECG acquisition hardware

The AD8232 is a single-lead heart-rate monitor front end that amplifies and filters the small biopotential signal picked up by three chest electrodes (right arm, left arm and right leg reference) [30]. Its analogue output is connected to GPIO 35 of the ESP32-WROOM-32, which belongs to the ADC1 block and can therefore be read while Wi-

Fi is active [31]. The two leads-off detection outputs, LO+ and LO-, are wired to GPIO 33 and GPIO 32. The ESP32 samples the ECG at 250 Hz and sends the samples to the hub with each telemetry packet. Figure 2.2 summarises this wiring.

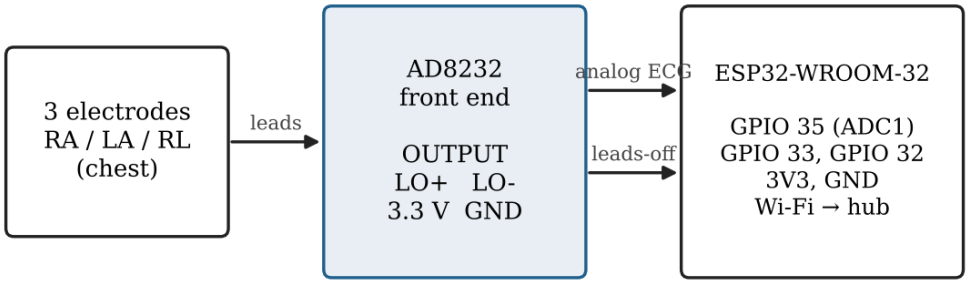


Figure 2.2: ECG acquisition hardware of the Smart Belt

Source: Author, based on the group's wiring.

2.2.3 Arrhythmia component pipeline

Inside the hub, the component runs the pipeline in Figure 2.3 each time new telemetry arrives, roughly twice per second. The most recent eight seconds of ECG are taken from a rolling buffer, R peaks are detected, features are computed for the most recent complete beat, and the beat is classified. In parallel, the heart rate reported by the glove is checked against the bradycardia and tachycardia rules and against the posture reported by the belt. Both results are then passed to the Master Risk Score.

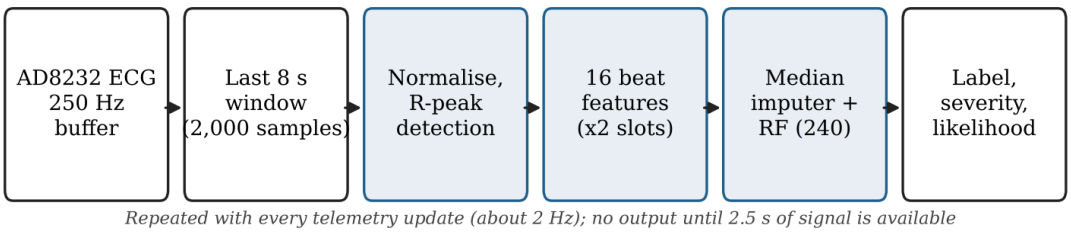


Figure 2.3: Processing pipeline of the arrhythmia component on the hub

Source: Author.

2.3 Dataset and Data Preparation

2.3.1 Dataset source

The classifier was developed with the public "ECG Arrhythmia Features Dataset" published by Biofic Analytics on Kaggle under the GPL 2 licence [34]. The group chose it because it already provides beat-level features labelled with the five AAMI classes, which matched the planned design, and because access to real clinical recordings with expert annotations was not available within the project timeline.

A closer look at the file changed how the results can be interpreted. The dataset card describes the data as designed to emulate real-world monitoring data, and the record identifiers confirm that they are synthetic: all 100 records are named synth_0000 to synth_0099, every record contains exactly 500 beats, and the class mix is almost identical from one record to the next. Real recordings never look like this. The study therefore uses 100 synthetic subject records (50,000 beats) and no real patient data. This was an important finding in itself, because it limits the clinical claims that can be made, and it is returned to in Section 3.3.

2.3.2 Class distribution

Each row is labelled with one of the five AAMI classes. Table 2.1 and Figure 2.4 show the distribution, computed directly from the dataset file. Normal beats make up about 60% of the data, so the problem is imbalanced, although much less so than MIT-BIH, where normal beats are close to 90% of the total [7].

Table 2.1: Beat class distribution of the dataset (50,000 beats, 100 records)

Class	Meaning	Beats	Share (%)
N	Normal beat	30,103	60.2
VEB	Ventricular ectopic beat	5,975	12.0
SVEB	Supraventricular ectopic beat	5,947	11.9
Q	Unclassifiable beat	4,070	8.1
F	Fusion beat	3,905	7.8

Source: Author's analysis of the dataset in [34].

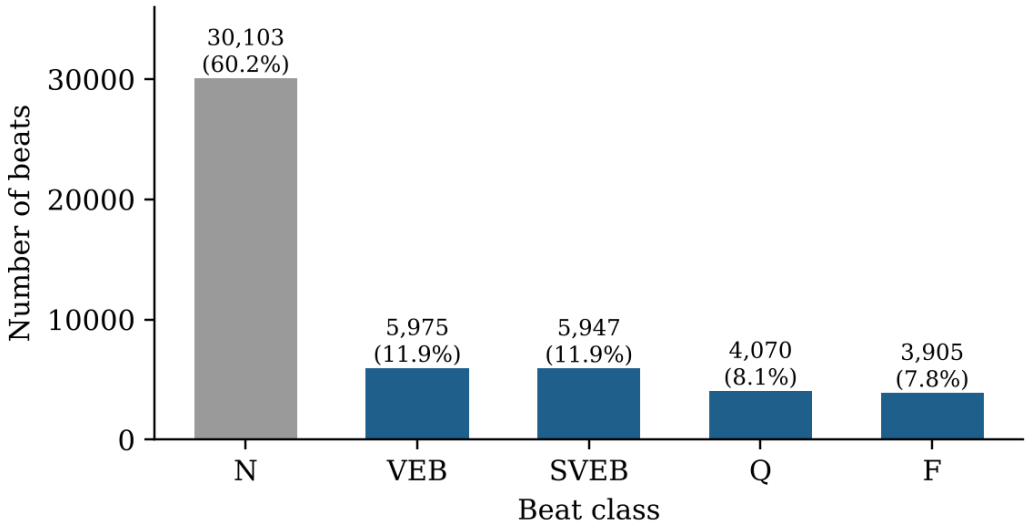


Figure 2.4: Number of beats in each class

Source: Author's analysis of the dataset in [34].

2.3.3 Features

Each beat is described by sixteen features, grouped in Table 2.2. The file stores them twice, with the prefixes 0_ (lead 0) and 1_ (lead 1). When the two sets were compared, they turned out to be identical in all 50,000 rows, so the dataset carries sixteen unique features rather than thirty-two. This fits the single-lead AD8232 used in SmartGlove. The classifier is still trained on all thirty-two columns, and at run time the single-lead features are copied into both slots so that the live input has the same layout as the training data. A second observation was that st_interval has the same value, 20 samples, in every row. At 250 Hz this corresponds to 0.08 s, which is consistent with the sampling rate of the AD8232 acquisition, but a constant column carries no information for the classifier.

Table 2.2: Feature groups used for beat classification (per lead)

Group	Features	Count
RR intervals	pre-RR, post-RR	2
Wave amplitudes	pPeak, tPeak, rPeak, sPeak, qPeak	5
Intervals	qrs_interval, pq_interval, qt_interval, st_interval	4
QRS morphology	qrs_morph0 to qrs_morph4	5
Total	| 16	

Source: Author's analysis of the dataset in [34].

2.3.4 Data checks and patient-wise split

Before training, the file was checked for missing values (none), duplicate rows (none), the set of class labels (five) and the record structure (100 records of 500 beats each). The records were then split 80% / 20% by record with scikit-learn's GroupShuffleSplit (test_size = 0.20, random_state = 42) [35], giving 80 training records (40,000 beats) and 20 test records (10,000 beats).

Splitting by record rather than by beat is the most important methodological choice in this chapter. Beats from the same person are very similar to each other. If they are shuffled into both the training and the test set, the classifier is partly tested on beats it has effectively already seen, and the results become optimistic. This is the reason de Chazal et al. introduced the inter-patient scheme [8], and the survey by Luz et al. shows how much reported accuracy drops when it is applied [9].

2.4 Live Feature Extraction

The classifier is only useful if the hub can compute the same features from the live signal. A function in backend/AI_models/arrhythmia/inference.py does this for the latest complete beat in the buffer. The procedure, illustrated in Figure 2.5, is as follows.

1. **Window and warm-up.** The last eight seconds of ECG (2,000 samples) are used. If fewer than 2.5 seconds are available, the component returns the status "ECG WARMING UP" without a prediction. A window with almost no amplitude variation, such as a disconnected electrode, returns no features.
2. **Normalisation.** The window is centred on its median and divided by its standard deviation, so that the features do not depend on the gain of the sensor. No separate band-pass filter is applied in the current version, because the AD8232 already has analogue filtering on board.
3. **R-peak detection.** Peaks are found with the SciPy peak finder [36] using a minimum spacing of 0.25 s (62 samples) and a prominence of 0.45 on the normalised signal. At least three peaks are required.

4. **Beat selection.** The second-to-last peak is classified, because it is the latest beat for which both the preceding RR interval (pre-RR) and the following RR interval (post-RR) are known.

5. **Wave points.** A window of 0.18 s on each side of the R peak is cut out. The Q and S points are taken as the minima before and after R, and the P and T peaks as the maxima in windows from 0.25 s to 0.08 s before R and from 0.08 s to 0.40 s after it.

6. **Feature vector.** The two RR intervals, five amplitudes, four intervals and five equally spaced samples across the beat window form the sixteen features, which are copied into both lead slots to give thirty-two inputs.

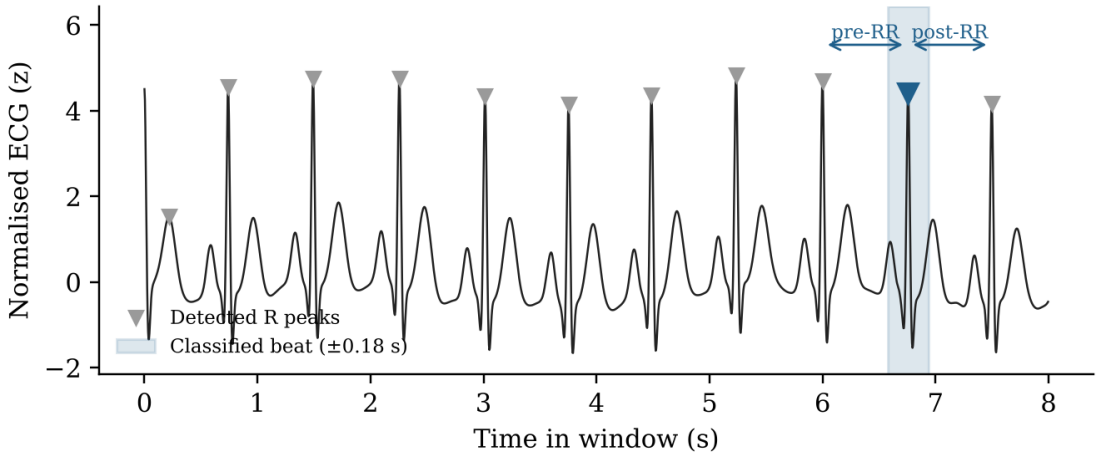


Figure 2.5: Example of an eight-second window with detected R peaks and the beat selected for classification

Source: Author, simulated ECG at 80 BPM [5], processed with the component's feature extraction.

Listing 2.1 shows the core of this routine. The full function is given in Appendix B.

Listing 2.1: R-peak detection and beat selection (excerpt of backend/AI_models/arrhythmia/inference.py)

```
normalized = (signal - np.median(signal)) / (np.std(signal) + 1e-8)
distance = max(1, int(0.25 * sampling_rate))
peaks, _ = find_peaks(normalized, distance=distance, prominence=0.45)
if len(peaks) < 3:
    return None
r = int(peaks[-2])                          # latest beat with both RR intervals
pre_rr = float(peaks[-2] - peaks[-3])
post_rr = float(peaks[-1] - peaks[-2])
```

2.5 Classifier Design and Training

2.5.1 Training procedure

The training function, train() in the arrhythmia module, builds a scikit-learn pipeline with two steps: a median imputer and a Random Forest classifier [20], [35]. The settings are listed in Table 2.3.

Table 2.3: Random Forest pipeline settings

Setting	Value	Reason
Imputer	Median, fitted on training data	Gives an output even when a feature cannot be computed
Number of trees	240	Chosen by tuning as a balance between variance and latency
Maximum depth	None	Trees grow until leaves are pure or reach the minimum size
Minimum samples per leaf	2	Limits over-specific leaves
Features per split	Square root of 32	Standard choice for classification forests [20]
Class weight	balanced_subsample	Re-weights classes in each bootstrap sample
Parallel jobs / seed	n_jobs = -1 / random_state = 42	Faster training and reproducible results

Source: Training script of the arrhythmia component.

Class imbalance was handled in two ways, following common practice for imbalanced learning [37]. Within the training set only, normal beats were randomly undersampled (seed 42) to at most three times the size of the largest abnormal class, while all abnormal beats were kept. The class weights then compensate for the remaining imbalance. The test set was left untouched, so that the evaluation reflects the natural class mix. The trained pipeline is stored with joblib as arrhythmia_model.joblib, which is 201.8 MB.

Listing 2.2: Patient-wise split, class balancing and pipeline (excerpt of the training script)

```
splitter = GroupShuffleSplit(n_splits=1, test_size=0.20, random_state=42)
train_index, test_index = next(splitter.split(x, y, groups=groups))
normal = np.flatnonzero(y_train.to_numpy() == "N")
abnormal = np.flatnonzero(y_train.to_numpy() != "N")
limit = min(len(normal), int(y_train.iloc[abnormal].value_counts().max() * 3))
selected = np.concatenate([rng.choice(normal, limit, replace=False),
abnormal])
```

```
pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("classifier", RandomForestClassifier(n_estimators=240,
        class_weight="balanced_subsample", min_samples_leaf=2,
        max_features="sqrt", n_jobs=-1, random_state=42)),
])
```

2.5.2 Choice of classifier

Random Forest was chosen at the start of the project for reasons that are common in the literature on tabular data [23]: it handles non-linear interactions between features without scaling, it is relatively robust to overfitting, it trains in parallel, and its feature importances can be shown to a clinician. Early trials with a support vector machine and XGBoost [22] did not give a clear reason to change. However, an untested assumption is a weak basis for a thesis, so a systematic comparison was carried out later in the project.

Seven classifiers were trained on the same sixteen features with the same median imputer, using five-fold cross-validation grouped by record so that no record appeared in both training and testing folds. Latency was measured per beat on a general-purpose CPU, so only the comparison between models is meaningful, not the absolute values. The results are given in Table 2.4.

Table 2.4: Comparison of candidate classifiers (five-fold cross-validation grouped by record)

Model	Accuracy (%)	Balanced acc. (%)	Macro F1 (%)	Latency (ms)	Size (KB)
Random Forest (240 trees, unrestricted)	90.6 ± 0.3	81.3 ± 0.5	85.3 ± 0.6	17.33	170,215
Random Forest (240 trees, depth 12, leaf 5)	84.5 ± 0.3	70.4 ± 0.8	74.2 ± 0.7	16.13	20,166
Decision tree	87.7 ± 0.4	83.2 ± 0.6	83.0 ± 0.5	0.35	592
Logistic regression	77.7 ± 0.5	59.4 ± 0.9	61.1 ± 0.6	0.43	3
k-nearest neighbours (k = 5)	87.9 ± 0.3	77.7 ± 0.5	81.7 ± 0.5	2.25	5,314
Histogram gradient boosting [21]	93.1 ± 0.4	86.2 ± 0.9	88.9 ± 0.8	5.92	1,763
Multilayer perceptron (64-32)	93.8 ± 0.1	88.6 ± 0.7	90.4 ± 0.5	0.72	116

Source: Author's experiments on the dataset in [34] with scikit-learn [35]. Values are mean ± standard deviation over five folds; the MLP was limited to 200 training iterations.

The table did not simply confirm the original choice, and it is reported here as it came out. Three points stand out. The linear model is clearly worse on balanced accuracy, which supports the need for a non-linear model. Histogram gradient boosting and the small multilayer perceptron are more accurate than the Random Forest and are also much smaller and faster. And a single decision tree is already close to the Random Forest on balanced accuracy at a fraction of the size, which suggests that the useful structure in these features is fairly simple. Limiting the depth of the forest reduces its size about eight times but costs eleven points of balanced accuracy.

The Random Forest was kept in the deployed system for the remainder of the project because it was already integrated and tested with the other components, its decisions can be explained through feature importances (Section 3.1.3), and its latency on the hub was within the real-time budget. Replacing it with gradient boosting, which keeps the explainability of trees while improving accuracy, is recorded as the first recommendation in Chapter 4.

2.5.3 Effect of class balancing

To check whether the balancing step helps, the deployed settings were trained with and without it on the same split. Table 2.5 shows that balancing gave the highest balanced accuracy and the best recall for the two rare classes, F and SVEB, at the cost of about half a point of overall accuracy. Since this is a single split, differences of one or two points should be read as indicative only.

Table 2.5: Effect of class balancing on the Random Forest (single split, random_state = 42)

Training setting	Train beats	Acc. (%)	Bal. acc. (%)	Macro F1 (%)	F recall (%)	SVEB recall (%)
Undersampled N + balanced weights (deployed)	30,390	90.6	83.9	86.0	54.3	77.0
No undersampling, balanced weights	40,000	90.9	82.1	85.6	47.4	73.0
No balancing	40,000	91.0	81.8	85.6	45.7	72.9

Source: Author's experiment on the dataset in [34], using the settings of the deployed training script.

The experiments in Tables 2.4 and 2.5 were run on the copy of the dataset file downloaded during the thesis work. On that file the deployed script reaches 90.6% accuracy and 83.9% balanced accuracy, whereas the group's recorded training run, which produced the deployed model evaluated in Chapter 3, reached 80.7% and 68.9%. The class counts of the two test sets also differ, so the two runs did not see the same data.Chapter 3 reports the deployed model's run, because that is the model the system actually uses.

2.6 Rhythm Rules, Posture-Aware Fusion and Risk Integration

2.6.1 Heart-rate rules

The beat classifier looks at one beat at a time and cannot see a slow or fast rhythm on its own. Two rules in backend/main.py cover this, using the heart rate reported with each packet:

- **Sustained bradycardia.** A heart rate above 0 and below 45 BPM increments a per-device counter, and any other reading resets it. When the counter reaches 10 consecutive readings, the alert SUSTAINED BRADYCARDIA (severity 2, trigger LOW_HR) is raised.
- **Tachycardia.** A heart rate above 140 BPM raises the alert TACHYCARDIA (severity 2, trigger HIGH_HR).

Both thresholds are deliberately more conservative than the clinical definitions of bradycardia (below 50 BPM) and tachycardia (above 100 BPM at rest) [32], [33]. The reason is that wearable heart-rate readings are noisy, and a screening device that alarms at every reading of 55 or 105 BPM would soon be ignored. The ten-reading condition for bradycardia stops a single dropped beat or a brief loss of contact from raising an emergency. The tachycardia rule has no such counter, because a fast rate is checked against posture instead.

2.6.2 Posture-aware fusion

The sensor fusion module (backend/AI_models/sensor_fusion.py) stores the latest posture reported by the belt for each device and compares the measured heart rate with

the range expected for that activity (Table 2.6). If the heart rate falls within the expected range, it is considered "activity-explained".

Table 2.6: Expected heart-rate range for each posture (ACTIVITY_EXPECTED)

Posture	Expected range (BPM)	Reading not explained when
Running	100 to 170	Above 170
Walking	80 to 120	Above 120
Standing	55 to 100	Above 100
Sitting	55 to 100	Above 100
Lying	50 to 90	Above 90

Source: ACTIVITY_EXPECTED in backend/AI_models/sensor_fusion.py.

Posture information gets less trustworthy as it gets older, so the module computes a fusion confidence that falls linearly with the age of the last belt update:

$$fusion\ confidence = max(0,\ 1 - age\ /\ 10\ s)$$

When the heart rate is activity-explained and the fusion confidence is above 0.5, that is, when the posture is less than five seconds old, the HIGH_HR alert is removed. Only the heart-rate alert is suppressed. An abnormal beat shape is not explained by exercise, so an ARRHYTHMIA alert from the classifier is kept regardless of posture. A heart rate of 150 BPM while lying down is never suppressed.

2.6.3 Screening output and risk integration

For each window the classifier returns class probabilities. The class with the highest probability determines the diagnosis label and its severity: N is NORMAL HEALTH (0), SVEB is SUPRAVENTRICULAR ECTOPIC BEAT (1), VEB is VENTRICULAR ECTOPIC BEAT (2), F is FUSION BEAT (2) and Q is UNCLASSIFIABLE ECG BEAT (1). The likelihood reported to the rest of the system is the probability that the beat is abnormal:

$$likelihood\ (\%) = (1 - P(N)) \times 100$$

If the screening is ready and its severity is 2 or more, an ARRHYTHMIA alert is placed at the top of the alert list. The Master Risk Score then combines all ready screening engines with the correlation engine developed by another group member:

$$Risk = round(0.7 \times max\ likelihood + 0.3 \times correlation\ score),\ \ Risk = 100\ for\ a\ fall\ or\ SOS$$

The correlation score measures how far the user's SpO2 and heart rate have moved from a rolling personal baseline, using z-scores. Taking the maximum likelihood, rather than the average, prevents one serious finding from being diluted by four healthy models. A score above 60 triggers the emergency alert. The complete decision flow is shown in Figure 2.6 and worked examples are given in Table 2.7.

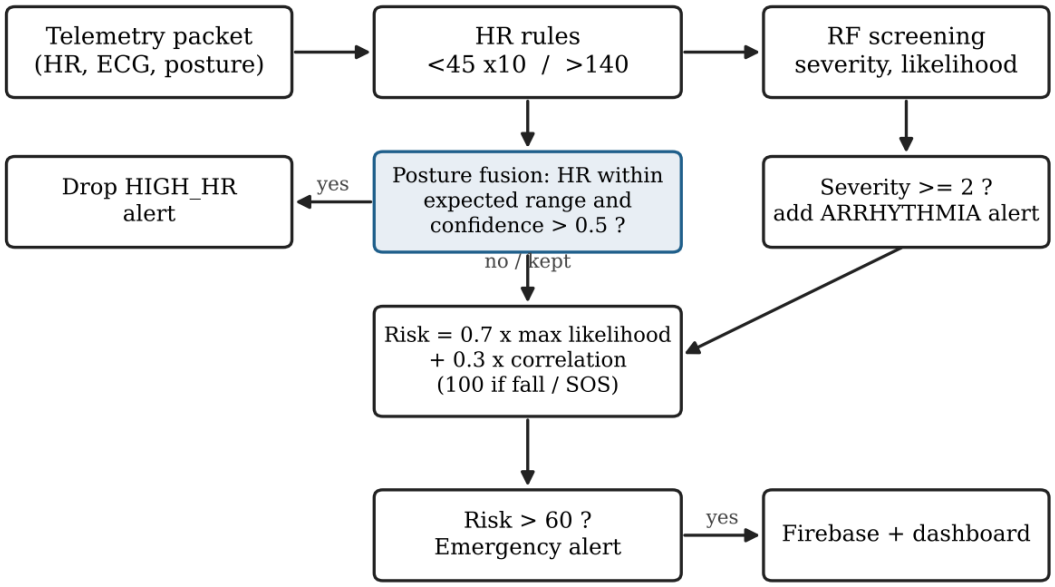


Figure 2.6: Alert decision flow combining heart-rate rules, posture fusion, the classifier and the Master Risk Score

Source: Author.

Table 2.7: Worked examples of the Master Risk Score

Case	Highest likelihood	Correlation score	Score	Emergency alert
Healthy, resting	10	10	10	No
Mild irregularity	50	20	41	No
Likely arrhythmia at rest	80	40	68	Yes
Fall or SOS gesture	any	any	100	Yes

Source: Author, using the formula in backend/main.py. The values are illustrative.

2.7 User Requirements and Stakeholder Analysis

2.7.1 Stakeholders

The requirements were shaped by the people who would use or depend on the system. Table 2.8 summarises them.

Table 2.8: Stakeholders of the arrhythmia component

Stakeholder	Role	Main needs
Elderly or at-risk users	Wear the glove and belt during daily life	Comfortable wear, no manual operation, few false alarms
Family members and carers	Receive alerts on the mobile app	Timely, clear alerts that say what happened
Doctors and nurses	Review alert history	Standard beat classes, honest accuracy figures, explainable output
Care homes and home-care services	Possible institutional buyers	Low cost per user, works without reliable internet
Other group members	Build the modules this component connects to	Stable output format, predictable timing

Source: Author.

2.7.2 Functional and non-functional requirements

Table 2.9 lists the requirements of the component and how each one is verified.

Table 2.9: Requirements of the arrhythmia component

ID	Requirement	Type	Verification
FR-01	Extract per-beat features from a single-lead ECG sampled at 250 Hz	Functional	Simulation tests (Section 3.1.5)
FR-02	Classify each beat as N, SVEB, VEB, F or Q	Functional	Held-out evaluation (Section 3.1.1)
FR-03	Report sustained bradycardia (< 45 BPM for 10 readings) and tachycardia (> 140 BPM)	Functional	Unit tests (Section 3.1.6)
FR-04	Suppress heart-rate alerts explained by the posture from the belt	Functional	Unit tests (Section 3.1.6)
FR-05	Pass severity and likelihood to the Master Risk Score	Functional	Unit and integration tests
FR-06	Report a warming-up status until enough ECG is available	Functional	Simulation tests

ID	Requirement	Type	Verification
NFR-01	Classify one window in under 4 ms on a CPU	Performance	Timing test (Section 3.1.4)
NFR-02	Run on the local hub without internet or a GPU	Deployment	Design review
NFR-03	Keep working when a feature is missing or noisy	Reliability	Edge-case tests
NFR-04	Give decisions that can be explained by measurable indicators	Explainability	Feature importance (Section 3.1.3)
NFR-05	Keep raw ECG on the hub	Privacy	Design review

Source: Author.

2.8 Commercialization Aspects of the Product

2.8.1 Target market and value proposition

The primary market for SmartGlove is elderly people and patients with known cardiac risk who live at home or in care homes, together with the family members who look after them. Sri Lanka's population is ageing quickly, many older people live with adult children who work during the day, and private hospitals are expanding home-care services. In this setting, families want reassurance that a problem will be noticed even when nobody is in the room.

The arrhythmia component contributes three things to the product's value proposition. It provides continuous beat-by-beat screening rather than a spot check; it raises alerts in real time on a local hub, so it keeps working when the internet connection drops; and it uses posture to avoid alarming during normal activity. All of this is built from components that cost a small fraction of a medical-grade recorder.

2.8.2 Competitive position

Table 2.10 compares SmartGlove's arrhythmia function with the alternatives available to a Sri Lankan family today.

Table 2.10: Competitive comparison for home arrhythmia monitoring

Option	Continuous	Real-time alert	Beat-level classes	Context-aware	Cost to user
24-hour Holter (hospital)	24 to 48 h	No	Yes (after review)	No	Test fee per use
ECG patch (e.g., Zio)	Up to 14 days	No	Yes (after review)	No	High, limited availability
Smartwatch with ECG	No (30 s on demand)	AF notification	No	No	High
Handheld ECG (e.g., KardiaMobile)	No (on demand)	No	No	No	Moderate
SmartGlove	Yes, while worn	Yes	Yes (5 classes)	Yes (posture)	Low (see Table 2.11)

Source: Author, based on [24], [26], [27].

SmartGlove does not compete with clinical recorders on diagnostic accuracy. Its niche is continuous screening and early warning between clinical visits, with a referral to a doctor when something abnormal is found.

2.8.3 Business model

Three revenue streams are proposed for the full SmartGlove product:

1. **Device kit sale** to families or institutions, covering the glove, belt and hub.
2. **Monthly subscription** for the carer mobile application, alert history and cloud synchronisation.
3. **Institutional licences** for care homes and home-care providers, with a dashboard covering many residents from one hub.

The arrhythmia component adds no running cost because classification runs locally. Cloud costs are limited to alert synchronisation, which fits within the free tier of Firebase at prototype scale.

2.8.4 Bill of materials

Table 2.11 lists the hardware used by the arrhythmia function. The hub is shared by all five screening models, so its cost is not specific to this component.

Table 2.11: Estimated bill of materials for the arrhythmia function

Item	Qty	Unit cost (LKR)	Total (LKR)	Purpose
ESP32-WROOM-32 development board	1	2,500	2,500	Belt microcontroller, Wi-Fi
AD8232 ECG module with lead cable	1	3,200	3,200	ECG front end
Disposable ECG electrodes (pack of 50)	1	1,500	1,500	Skin contact
Raspberry Pi 4 Model B, 4 GB (shared hub)	1	28,000	28,000	Runs Flask backend and models
microSD card 32 GB and power supply	1 set	5,000	5,000	Hub storage and power
Chest strap, enclosure, wiring	1 set	3,500	3,500	Mounting and assembly
Software (Python, scikit-learn, Flask, Firebase free tier)	| 0	0	Open source / free tier	
Total	|	43,700		

The 201.8 MB model file is one reason a Raspberry Pi with at least 4 GB of memory is listed. A gradient-boosted model of about 2 MB (Table 2.4) would allow a cheaper hub.

2.8.5 Regulation, intellectual property and sustainability

Software that analyses ECG to detect arrhythmia is classed as a medical device in most jurisdictions, and is described internationally as software as a medical device [38]. In Sri Lanka, medical devices fall under the National Medicines Regulatory Authority Act, No. 5 of 2015 [39]. Before SmartGlove could be sold with an arrhythmia claim, the group would need a risk management process following ISO 14971 [40], a software life cycle following IEC 62304 [41], and testing against the particular standard for ambulatory ECG systems, IEC 60601-2-47 [42]. Because the system handles health data and location, the Personal Data Protection Act, No. 9 of 2022 also applies [43]. The local processing design already helps here, since raw ECG stays on the hub.

The main intellectual property of the component is the integration of beat classification with posture-aware rules and risk scoring on a low-cost hub. The underlying libraries

are open source and permissively licensed, while the dataset is distributed under GPL 2 [34], which should be reviewed before commercial use of a model trained on it.

The product supports United Nations Sustainable Development Goal 3, good health and well-being, and in particular target 3.4, which aims to reduce premature mortality from noncommunicable diseases by one third [44]. By offering low-cost monitoring that does not depend on internet coverage, it also relates to Goal 10 on reducing inequalities in access to care.

2.9 Implementation

2.9.1 Technology stack

The technologies used by the component are summarised in Table 2.12.

Table 2.12: Technology stack of the arrhythmia component

Layer	Technology	Role
Microcontroller	ESP32-WROOM-32	Samples the ECG at 250 Hz and sends telemetry
ECG sensor	AD8232	Analogue front end; output on GPIO 35, leads-off on GPIO 32 / 33
Transport	HTTP POST over local Wi-Fi	Devices push JSON packets to the hub
Hub	Local PC or Raspberry Pi	Runs the backend and the models
Backend	Python 3, Flask 3.0 or later	Receives telemetry, runs rules, models and fusion
Machine learning	scikit-learn 1.3 or later, joblib, NumPy, SciPy	Pipeline, model storage, R-peak detection
Cloud	Firebase Realtime Database (firebase-admin 6.5 or later)	Synchronises alerts for the mobile app
Dashboard	Flask templates, HTML, JavaScript, Server-Sent Events	Live vitals and alerts

Source: Author and the project repository.

2.9.2 Component implementation

Training. The train() function loads the dataset, applies the patient-wise split, balances the training set and saves the fitted pipeline (Section 2.5).

Inference. The inference module is loaded once when the backend starts. Its predict() method receives the ECG buffer with each telemetry update, extracts features from the latest eight seconds, calls predict_proba() and returns a dictionary with the diagnosis, severity, likelihood and status. While less than 2.5 s of signal is available it returns ECG WARMING UP with the status warming_up.

Heart-rate rules. Listing 2.3 shows the rule function in the hub.

Listing 2.3: Heart-rate rules in the hub (excerpt of backend/main.py)

```
def evaluate_rules(self, vitals, device_id):
    hr = vitals.get("heart_rate")
    if hr is not None:
        if 0 < hr < 45:
            self.bradycardia_counts[device_id] += 1
            if self.bradycardia_counts[device_id] >= 10:
                alerts.append({"diagnosis": "SUSTAINED BRADYCARDIA",
                               "severity": 2, "trigger": "LOW_HR"})
        else:
            self.bradycardia_counts[device_id] = 0
            if hr > 140:
                alerts.append({"diagnosis": "TACHYCARDIA",
                               "severity": 2, "trigger": "HIGH_HR"})
```

Fusion and risk. The fusion module looks up the expected range for the current posture, computes the fusion confidence and flags whether the heart rate is activity-explained. The hub then removes HIGH_HR alerts where appropriate, inserts the ARRHYTHMIA alert when the severity is 2 or more, and computes the Master Risk Score (Listing 2.4).

Listing 2.4: Fusion-based suppression and Master Risk Score (excerpt of backend/main.py)

```
if suppress_hr_fusion:
    rules_alerts = [a for a in rules_alerts if a.get("trigger") != "HIGH_HR"]
arr = screenings.get("arrhythmia", {})
if arr.get("status") == "ready" and arr.get("severity", 0) >= 2:
    rules_alerts.insert(0, {"diagnosis": f"ARRHYTHMIA:
{arr.get('diagnosis')}",
                            "severity": arr.get("severity", 2)})
risk_score = 100 if is_emergency else 0
if not is_emergency and ready_likelihoods:
    risk_score = int(round(0.7 * max(ready_likelihoods) + 0.3 *
correlation_score))
```

2.9.3 Integration with the other components

On the hub, each telemetry update is processed in a fixed order. The screening models, including the arrhythmia classifier, process the latest data; the fusion module compares the wrist vitals with the belt posture; the heart-rate rules are applied and activity-explained HIGH_HR alerts are removed; an ARRHYTHMIA alert is added if needed;

and the Master Risk Score is calculated. When the score exceeds 60, or when a fall or SOS gesture is detected, the alert is written to Firebase and shown on the dashboard. The component therefore depends on the belt for the ECG and posture, on the glove for the heart rate, and on the correlation and alert modules of other members for the final decision.

2.10 Testing Strategy

Testing was planned at four levels, as summarised in Table 2.13. The detailed results are reported in Chapter 3 and the full test case matrix is given in Appendix A.

Table 2.13: Testing levels used for the component

Level	What is tested	Method
Model evaluation	Classification of held-out beats	Patient-wise test set of 20 records; standard metrics
Feature extraction	R-peak detection and feature values on live-like signals	Simulated ECG at 45 to 170 BPM with noise, baseline wander and 50 Hz interference
Unit tests	Heart-rate rules, fusion confidence, suppression, label mapping, risk score	pytest test suite (17 tests)
Integration and scenario	Alert flow from device to Firebase and dashboard	Scripted scenarios on the running system

Source: Author.

Simulated ECG was generated with the ECGSYN model [45] as implemented in NeuroKit2 [5]. This model produces realistic P-QRS-T morphology and heart-rate variability at a chosen rate and noise level. For each of six heart rates (45, 60, 75, 100, 140 and 170 BPM) and three noise levels, twenty twelve-second signals were generated, and the eight-second window was moved forward every 0.5 s, as it would be with live telemetry. A window counted as correct when both pre-RR and post-RR were within 10% of the true beat interval.

Classification performance is reported with accuracy, per-class precision, recall and F1, balanced accuracy (the mean of per-class recalls, which gives each class equal weight [46]), macro F1 and the confusion matrix. Accuracy alone is misleading here

because normal beats dominate the test set. For a screening tool, recall on the ectopic classes matters most, since a missed abnormal beat costs more than a false alarm.

2.11 Ethics, Privacy and Responsible AI

The offline experiments used a public synthetic dataset with no personal data [34], and the live tests used simulated signals, so no human participants were involved in the evaluation reported here.

Raw ECG is processed on the hub and is not sent to the internet; only alerts, the related readings and the location needed for an emergency are synchronised to Firebase. Access to the database is controlled through the Firebase service account used by the backend.

SmartGlove is a screening aid, not a diagnostic device. Several design choices support responsible use: an explainable tree model, median imputation so that a missing feature does not crash the system, debounce logic for bradycardia, posture-aware suppression of exercise-related alerts, and an honest report of the weaknesses found during testing (Section 3.3).

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Results

3.1.1 Classification of held-out records

The deployed model was evaluated on the 20 held-out records, which contain 10,000 beats: 5,995 N, 1,219 VEB, 1,180 SVEB, 837 F and 769 Q. None of these records was seen during training. Table 3.1 gives the overall results.

Table 3.1: Overall results of the deployed model on the held-out test records

Measure	Result
Accuracy	80.71%
Balanced accuracy	68.90%
Macro F1	72.61%
Test beats	10,000 (20 records)

Source: Training and evaluation output of the deployed model.

The gap of almost twelve points between accuracy and balanced accuracy shows that the model does much better on some classes than on others. The per-class results in Table 3.2 and the confusion matrix in Figure 3.1 show where the difference comes from.

Table 3.2: Results for each beat class

Class	Test beats	Precision (%)	Recall (%)	F1 (%)
N	5,995	80.4	94.0	86.7
SVEB	1,180	67.5	36.9	47.7
VEB	1,219	78.3	73.3	75.7
F	837	77.3	40.4	53.1
Q	769	99.9	100.0	99.9

Source: Author's calculation from the confusion matrix in Figure 3.1. Values may differ by 0.1 from the training log because of rounding.

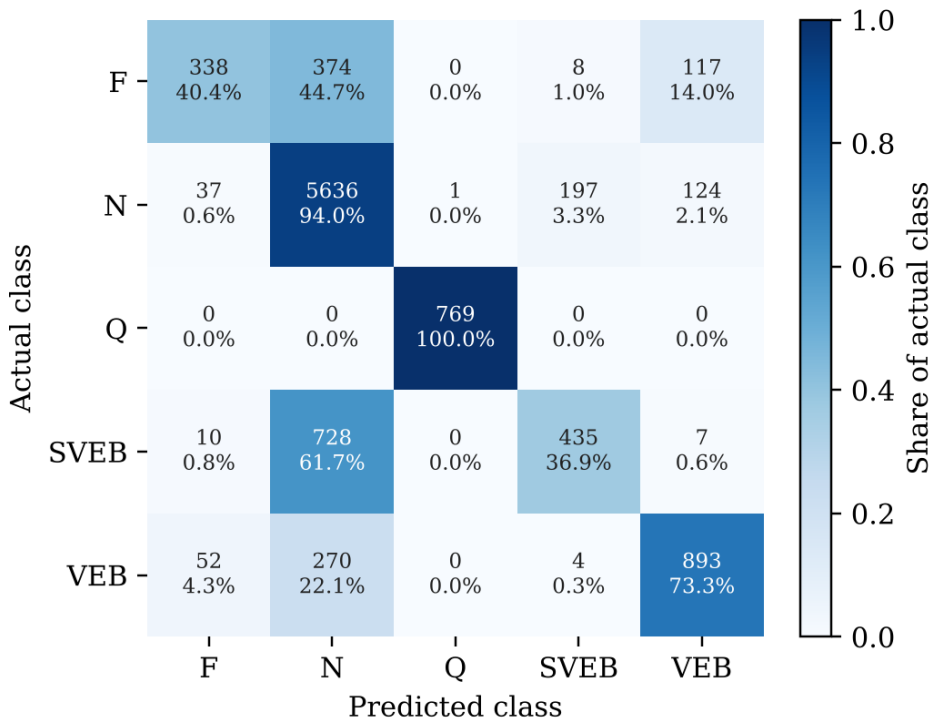


Figure 3.1: Confusion matrix on the 10,000 test beats (rows: actual class, columns: predicted class)

Source: Evaluation output of the deployed model; plotted by the author.

Normal beats are recognised well, with a recall of 94.0%, and every Q beat is found. Ventricular ectopic beats reach a recall of 73.3% and an F1 of 75.7%. The two weak classes are SVEB and F. Of the 1,180 SVEB beats, 728 (61.7%) were labelled normal, and of the 837 F beats, 374 (44.7%) were labelled normal and a further 117 (14.0%) were labelled VEB. Figure 3.2 shows the same pattern: the precision of SVEB and F is reasonable, but their recall is low, which means the model rarely calls a beat SVEB or F without reason but misses most of them.

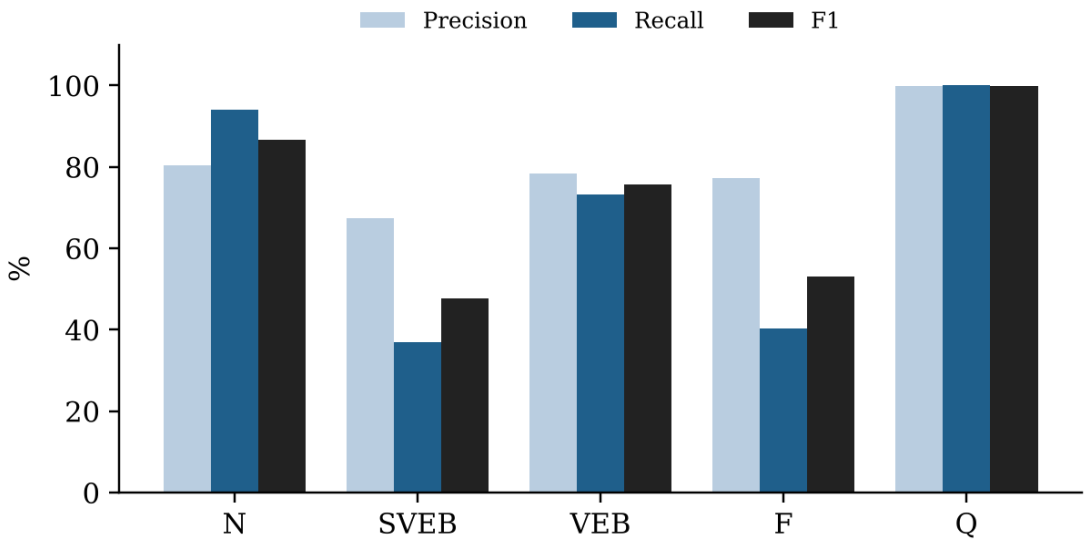


Figure 3.2: Precision, recall and F1 for each class

Source: Author.

3.1.2 Normal-versus-abnormal view

The Master Risk Score does not use the predicted class directly. It uses the abnormal probability, one minus the probability of N. The results can therefore also be read as a binary decision in which every class other than N counts as abnormal. Table 3.3 gives this view.

Table 3.3: Normal-versus-abnormal view of the results

Measure	Value	Meaning
Sensitivity	65.7%	2,633 of the 4,005 abnormal beats were not labelled N
Specificity	94.0%	5,636 of the 5,995 normal beats were labelled N
Precision of an abnormal call	88.0%	2,633 of the 2,992 beats not labelled N were truly abnormal

Source: Author's calculation from the confusion matrix in Figure 3.1.

Of the 3,236 ectopic beats (SVEB, VEB and F), 1,372 (42.4%) were labelled normal. In screening terms, the model is specific but not sensitive enough: when it raises a flag it is usually right, but it misses a considerable share of abnormal beats.

3.1.3 Feature importance

Figure 3.3 shows the eight most important features of the Random Forest in the reference experiment, measured by the mean decrease in impurity. The two RR intervals and the Q and S wave amplitudes are the strongest. This agrees with clinical reasoning and with earlier studies [9], [8]: beat timing reveals premature beats, and the depth of the Q and S waves reflects the shape of the QRS complex.

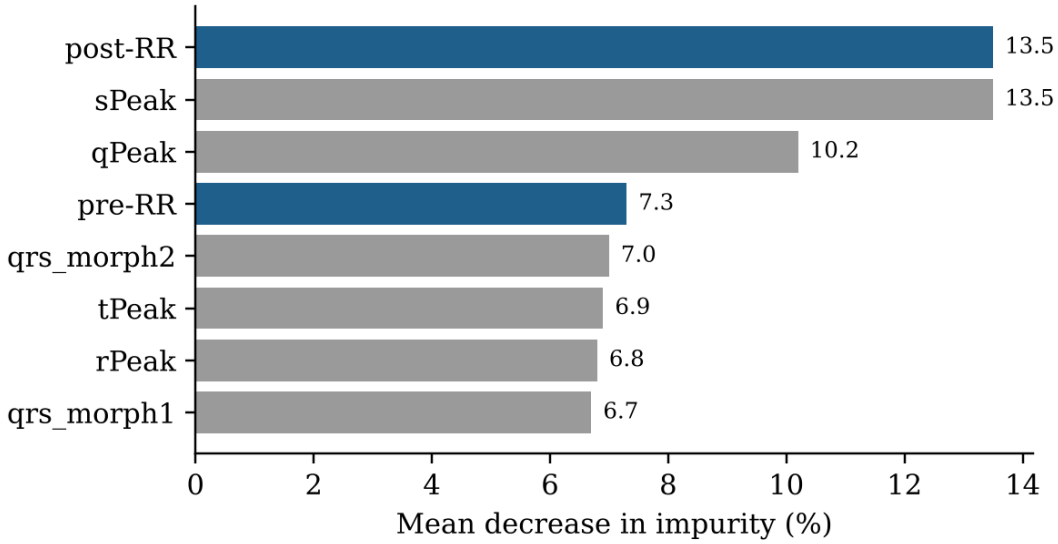


Figure 3.3: Eight most important features of the Random Forest (RR-interval features highlighted)

Source: Author's experiment on the dataset in [34].

3.1.4 Timing and resource use

Table 3.4 summarises the technical results. The classifier call takes 3.4 ms per window on the hub CPU, which meets the target of under 4 ms (NFR-01). The feature extraction, which was not included in that figure, was timed separately by the author over 2,000 runs on a two-core Intel Xeon 2.8 GHz machine: it took 0.33 ms on average (median 0.30 ms, 95th percentile 0.50 ms). Even together, the two steps use well under 1% of the 500 ms available between telemetry updates.

Table 3.4: Technical results

Item	Result	Note
Classifier inference	3.4 ms per window	Hub CPU; predict_proba() only

Item	Result	Note
Feature extraction	0.33 ms mean (0.50 ms 95th percentile)	Author's measurement, 2,000 runs
ECG window	8 s (2,000 samples)	250 Hz; first output after 2.5 s
Update rate	About 2 Hz	Buffer processed with every packet
Model inputs	32 columns	16 unique features in two lead slots
Model file	201.8 MB	arrhythmia_model.joblib
Internet needed	No	Runs on the local hub

Source: Author and the group's measurements.

It should be noted that the per-beat latency of the unrestricted Random Forest in Table 2.4 (17.3 ms) is higher than the 3.4 ms measured on the hub. The two figures were measured on different machines and in different ways: the comparison in Table 2.4 timed single-beat predictions on a shared general-purpose CPU, including scikit-learn's per-call overhead, while the hub figure was measured on the group's own computer.

3.1.5 Feature extraction on simulated ECG

The classifier was trained on features computed by the dataset authors, but in the running system the features come from the component's own extraction routine. To see whether this routine behaves correctly on live-like signals, it was run on simulated ECG as described in Section 2.10. The results were the most surprising part of the evaluation.

Table 3.5: Share of eight-second windows with correct pre-RR and post-RR (noise level 0.1)

Heart rate (BPM)	45	60	75	100	140	170
Current detector (%)	0.0	0.0	73.3	81.1	67.8	81.1
Revised detector (%)	100	100	100	100	100	100

Source: Author's simulation (360 windows per heart rate); results for noise levels 0 and 0.2 were similar.

Table 3.5 and Figure 3.4 summarise the results. At 45 and 60 BPM, the current detector never produced correct RR intervals, and Figure 3.5 shows why. At slow heart rates the P and T waves have more room between beats, and with a fixed prominence of

0.45 on the normalised signal, the peak finder accepts one of these smaller waves as an extra beat. The 60 BPM window in that figure contains eight real beats, but sixteen peaks were detected. Every pre-RR and post-RR value is then roughly halved, and the classifier receives features that look like premature beats. At normal and fast rates the RR values were mostly correct, but in 19% to 32% of windows the last peak in the window was a partly cut-off beat at the edge, which gives a wrong post-RR.

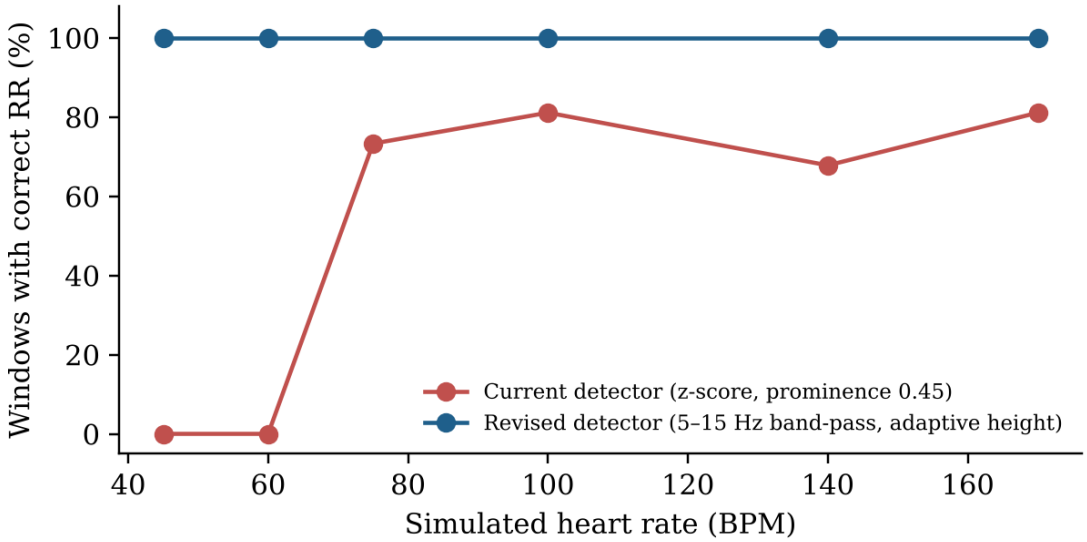


Figure 3.4: Windows with correct RR intervals against heart rate for the current and revised R-peak detectors

Source: Author's simulation.

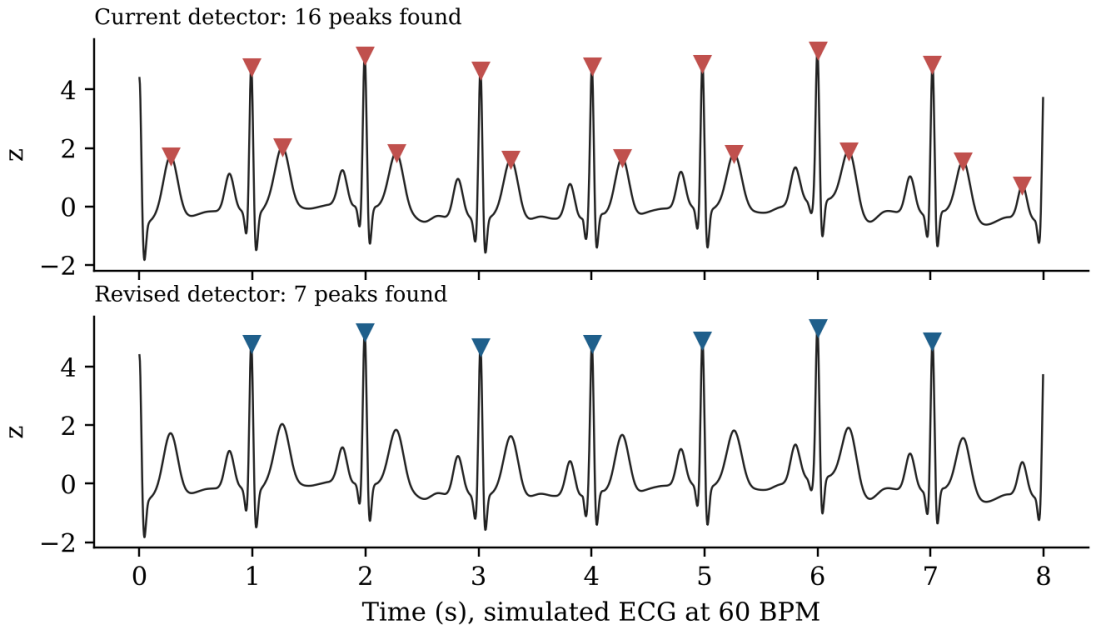


Figure 3.5: Detected peaks in the same 60 BPM window with the current and revised detectors

Source: Author's simulation.

A revised detector was written to test whether this could be fixed simply. It band-pass filters the normalised signal between 5 and 15 Hz, which keeps the steep QRS complex and suppresses the slower P and T waves, as in the classic Pan-Tompkins approach [47]. It squares the result, accepts peaks above 30% of the 99th percentile of the window, and ignores peaks within 0.15 s of the window edges. The rest of the feature extraction is unchanged. With this change, every window at every heart rate and noise level gave correct RR intervals. Table 3.6 shows that the revised detector also coped with baseline wander and 50 Hz mains interference, which caused the current detector to fail completely at 60 BPM.

Table 3.6: Windows with correct RR intervals under baseline wander (0.3 Hz) and 50 Hz interference

Heart rate (BPM)	Current detector (%)	Revised detector (%)
60	0.0	100
75	32.8	100
100	83.3	100

Source: Author's simulation.

Four edge cases were also checked. A buffer of only two seconds, a flat signal (as with a disconnected electrode) and a saturated signal at the ADC limit all correctly returned no features. However, a window of pure random noise did produce a full feature vector, which the classifier would then label. The routine has no check on signal quality, and the leads-off outputs of the AD8232, which are wired to GPIO 32 and 33, are not used to reject such windows.

3.1.6 Unit tests of the hub logic

The rule, fusion, label mapping and risk functions were tested in isolation with a pytest suite of 17 tests (Appendix A). All 17 passed. Table 3.7 groups them.

Table 3.7: Unit test results for the hub logic

Area	Tests	Examples	Result
Bradycardia rule	5	No alert after 9 readings, alert on the 10th, reset by a normal reading, separate counters per device, HR of 0 ignored	All passed
Tachycardia rule	2	140 BPM no alert, 141 BPM alert; missing HR handled	All passed
Posture fusion	4	Running at 150 BPM suppressed; lying at 150 BPM kept; posture older than 5 s not used; confidence 1.0, 0.5, 0.0 at 0, 5, 12 s	All passed
Screening output	2	VEB maps to severity 2 and likelihood 90%; N maps to severity 0	All passed
Risk score	4	Worked examples of Table 2.7; emergency gives 100; maximum engine used	All passed

Source: Author's test run (pytest).

Writing these tests exposed one detail of the risk score that is easy to miss. With a correlation score of zero, the classifier alone needs an abnormal likelihood of at least 87% to push the score above 60, because 0.7 × 86 rounds to 60. In one test, 0.7 × 85 gave 59 rather than the expected 60, because 59.5 is stored in floating point as 59.4999. This does not affect the ARRHYTHMIA alert, which is raised separately whenever the severity is 2 or more, but it means that the classifier on its own rarely triggers the emergency path.

3.1.7 System test cases

Table 3.8 summarises the twenty test cases defined for the component. Test cases TC-01 to TC-17 were executed by the author as simulation, timing or unit tests, except TC-09, which uses the group's timing measurement. TC-18 to TC-20 depend on the full hardware set-up and are reported from the group's integration testing.

Table 3.8: Summary of test case execution (TC-01 to TC-20)

TC	Test case	Method	Expected result	Status
TC-01	Buffer shorter than 2.5 s	Simulation	No features (warming up)	Passed
TC-02	Flat signal (electrode off)	Simulation	No features	Passed
TC-03	Saturated ADC signal	Simulation	No features	Passed
TC-04	RR intervals at 75 to 170 BPM	Simulation	Correct pre-RR and post-RR	Partly passed (68% to 81%)
TC-05	RR intervals at 45 and 60 BPM	Simulation	Correct pre-RR and post-RR	Failed (0%); revised detector passed
TC-06	Baseline wander and 50 Hz noise	Simulation	Correct RR intervals	Failed at 60 and 75 BPM; revised detector passed
TC-07	Pure noise input	Simulation	Window rejected	Failed (features returned)
TC-08	Feature extraction time	Timing	Under 4 ms	Passed (0.33 ms)
TC-09	Classifier inference time	Timing	Under 4 ms	Passed (3.4 ms)
TC-10	Bradycardia, 9 readings	Unit test	No alert	Passed
TC-11	Bradycardia, 10 readings	Unit test	SUSTAINED BRADYCARDIA	Passed
TC-12	Bradycardia counter reset and per device	Unit test	Counter resets; devices independent	Passed
TC-13	Tachycardia boundary (140 / 141 BPM)	Unit test	Alert only above 140	Passed
TC-14	Running at 150 BPM, fresh posture	Unit test	HIGH_HR suppressed	Passed
TC-15	Lying at 150 BPM; stale posture	Unit test	HIGH_HR kept	Passed
TC-16	Label, severity and likelihood mapping	Unit test	As in Section 2.6.3	Passed
TC-17	Risk score examples and emergency	Unit test	As in Table 2.7; 100 for emergency	Passed
TC-18	ARRHYTHMIA alert reaches Firebase and dashboard	Integration	Alert visible on app and dashboard	Passed
TC-19	End-to-end time from packet to alert	Integration	Under 1 s	Passed

TC	Test case	Method	Expected result	Status
TC-20	Exercise with belt worn (jogging)	Scenario	No TACHYCARDIA emergency	Passed

Source: Author; TC-18 to TC-20 from the group's integration tests.

3.2 Research Findings

The results answer the three research questions as follows.

1. **RQ1, classification accuracy.** On synthetic held-out records, the Random Forest reached 80.7% accuracy and 68.9% balanced accuracy. It recognised normal, unclassifiable and ventricular ectopic beats well but missed most supraventricular and fusion beats. Gradient boosting and a small neural network performed better in the comparison, with balanced accuracies of 86.2% and 88.6%, while a linear model performed clearly worse.
2. **RQ2, real-time operation.** Classification (3.4 ms) and feature extraction (0.33 ms) together take a small fraction of the 500 ms between updates, so the component runs comfortably in real time on CPU-only hardware. The large model file (201.8 MB) is the main resource cost, and it rules out running the classifier on the ESP32 itself.
3. **RQ3, reliability of the live pipeline.** The current R-peak detector is not reliable at resting heart rates below about 70 BPM, where it counts P or T waves as beats, and it gives a wrong post-RR in roughly a quarter of windows at higher rates. A band-pass detector with an adaptive threshold removed these errors in all simulated conditions. The pipeline also lacks a signal quality check.
4. **Rules and fusion.** The bradycardia, tachycardia and posture fusion logic behaved as designed in all unit tests, including the boundary cases. The risk score formula means the classifier alone seldom triggers the emergency path, which should be a conscious design decision rather than a side effect.

3.3 Discussion

3.3.1 Interpretation of the classification results

The pattern of errors makes sense when the features are considered. VEBs have a wide, deep QRS complex, which shows up clearly in the Q and S amplitudes and in the morphology samples, so the model finds most of them. SVEBs keep an almost normal QRS shape and differ mainly in timing, so the model has to rely on the RR features, and when the timing difference is small they look normal. Fusion beats are, by definition, a mix of a normal and a ventricular beat, which explains why they were confused with both. The same difficulty with SVEB appears throughout the literature [9], [15].

Table 3.9 places the results next to published inter-patient results on MIT-BIH. The comparison has to be read carefully. The published studies used real clinical recordings at 360 Hz, while this study used synthetic records, so the numbers are not directly comparable. What the table does show is that the precision for SVEB in this study is higher than in the published work, while its recall is much lower. The model is cautious with SVEB, where most published models are over-eager.

Table 3.9: Results of this study alongside published inter-patient results on MIT-BIH

Study	Data	SVEB Se (%)	SVEB +P (%)	VEB Se (%)	VEB +P (%)
de Chazal et al. [8]	MIT-BIH	75.9	38.5	77.7	81.9
Llamedo and Martínez [10]	MIT-BIH	76.5	41.3	82.9	88.0
Zhang et al. [12]	MIT-BIH	79.1	36.0	85.5	92.8
Huang et al. [15]	MIT-BIH	91.1	42.2	93.9	90.9
This study (deployed RF)	Synthetic	36.9	67.5	73.3	78.3

Source: Published values as compiled in [15]; this study from Table 3.2. Se: sensitivity (recall); +P: positive predictivity (precision).

The balancing experiment in Table 2.5 suggests one direction: balancing raised SVEB recall, and a threshold on the abnormal probability lower than the default arg-max decision would push recall further at some cost in precision. For a screening tool this trade-off is usually worth making.

3.3.2 Synthetic data and the gap to the live signal

The single biggest limitation of this work is that the classifier has never seen a real heartbeat. The dataset is synthetic, its records are unrealistically uniform, and its features were produced by a pipeline the group did not write. The simulation tests in Section 3.1.5 show why this matters. Even if the classifier were perfect on the dataset, it would receive wrong RR intervals from the live detector for a typical resting adult, and the RR features are among its most important inputs (Figure 3.3). There may also be differences in scale. For example, the live amplitudes are z-scores, and it has not been confirmed that the dataset's amplitudes were normalised the same way.

This finding is uncomfortable but useful. It shows that an accuracy figure on a dataset is only one link in a chain, and that the feature extraction deserves as much testing as the model. It also gives a clear and inexpensive next step: record ECG from volunteers with the actual AD8232 hardware, compare the distribution of each live feature with the training data, and label a sample of beats with the help of a clinician.

3.3.3 Design of the rules and the risk score

The heart-rate rules and the posture fusion worked as intended in testing, and the idea of suppressing only the heart-rate alert while keeping the beat-shape alert proved sound. Two weaknesses remain. The 45 and 140 BPM thresholds were chosen to reduce false alarms rather than derived from data, and they are more permissive than clinical definitions [32], [33]; a user resting at 110 BPM would not be flagged at all. The 0.7 and 0.3 weights of the risk score and the cut-off of 60 were also set by judgement. A small sensitivity analysis on recorded scenarios would put these values on a firmer footing.

3.3.4 Threats to validity

- **Internal validity.** The deployed model and the comparison experiments were trained on different versions of the dataset file (Section 2.5.3). The balancing experiment used a single split.

- **External validity.** All classification results come from synthetic data, and all live tests from simulated ECG. Neither can stand in for real patients, real electrode placement or real motion.
- **Construct validity.** Per-beat metrics do not measure what a user experiences, which is alerts over time. A model that misses some SVEBs may still flag a patient with frequent SVEBs within a minute.
- **Measurement.** The latency figures in Tables 2.4 and 3.4 were measured on different machines (Section 3.1.4).

3.4 Summary of Individual Contribution (IT22081148)

As the member of Group R26-IT-108 responsible for the arrhythmia component, Abeysinghe S.A.M.M. (IT22081148) carried out the following work:

1. **Dataset analysis.** Examined the ECG feature dataset and identified that it is synthetic, that its two lead sets are identical, and that one feature is constant; designed the patient-wise split.
2. **Model development.** Built the training pipeline with median imputation, class balancing and a 240-tree Random Forest; produced the evaluation metrics and the confusion matrix analysis.
3. **Live feature extraction and inference.** Implemented the R-peak detection, beat selection and feature computation on the hub, and the inference module that returns the diagnosis, severity, likelihood and status.
4. **Hub logic.** Implemented the bradycardia and tachycardia rules, the posture-aware suppression using the belt data, and the hand-over of the arrhythmia result to the Master Risk Score.
5. **Evaluation.** Ran the classifier comparison and the balancing experiment, the simulation tests of the feature extraction, the timing measurements and the unit tests, and proposed the revised R-peak detector.

CHAPTER 4: CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusions

This research set out to build an arrhythmia screening component for the SmartGlove wearable system that classifies heartbeats from a single-lead ECG on a local hub, interprets the heart rate in the light of the user's posture, and contributes to a shared risk score without relying on the cloud. That component was built, integrated with the rest of the system and evaluated.

The classifier assigns each beat to one of the five AAMI classes from sixteen engineered features. On 10,000 beats from 20 held-out synthetic records it reached 80.7% accuracy and 68.9% balanced accuracy. It is dependable for normal and ventricular ectopic beats and weak for supraventricular and fusion beats, a pattern that matches the wider literature. The classifier and its feature extraction together run in under 4 ms on CPU-only hardware, so real-time operation on a low-cost hub is clearly feasible. The heart-rate rules and the posture-aware fusion behaved correctly in every unit test, and suppressing only the heart-rate alert during activity, while keeping the beat-shape alert, proved a sensible way to reduce false alarms without hiding abnormal beats.

The evaluation also produced two findings that were not expected at the start. First, when the Random Forest was compared systematically with other classifiers, gradient boosting and a small neural network turned out to be more accurate and much smaller. Second, simulation tests showed that the live R-peak detector double-counts beats at resting heart rates below about 70 BPM, and a band-pass detector with an adaptive threshold fixed this in every simulated case. Both findings matter more than a single accuracy figure, because they show where the system would fail first in real use and what it would take to fix it.

Table 4.1 summarises how far each specific objective was achieved.

Table 4.1: Achievement of the specific objectives

Objective	Outcome	Status
1. Analyse the dataset and design a leakage-free split	Dataset found to be synthetic; patient-wise split applied	Achieved
2. Implement live feature extraction	Implemented; weakness at low heart rates found and a fix demonstrated	Partly achieved
3. Train and compare classifiers	Random Forest trained; seven models compared	Achieved
4. Heart-rate rules and posture fusion	Implemented and unit-tested	Achieved
5. Integrate with the Master Risk Score	Implemented; behaviour of the formula analysed	Achieved
6. Evaluate the component	Model, timing, simulation and unit tests completed	Mostly achieved

Source: Author.

In short, the work shows that beat-level arrhythmia screening with posture-aware alerting is practical on inexpensive edge hardware, and it gives an honest account of what must change before such a component can be trusted with real patients.

4.2 Recommendations and Future Directions

The following steps are recommended, in order of priority.

1. **Adopt the revised R-peak detector.** It is a small change to one function and removes the largest error found in this study. It should be tested on real recordings before deployment.
2. **Collect and label real ECG from the actual hardware.** Even a few hours of AD8232 recordings from volunteers, with a sample of beats labelled by a clinician, would allow the live features to be compared with the training data and the model to be fine-tuned on realistic signals. Public clinical databases such as MIT-BIH [7], resampled to 250 Hz and passed through the component's own feature extraction, would provide a second, independent test set.
3. **Replace the Random Forest with gradient boosting.** Histogram gradient boosting gave higher balanced accuracy in the comparison, keeps the explainability of trees, and is about one hundredth of the size, which would allow a cheaper hub.

4. **Add a signal quality check.** Use the AD8232 leads-off outputs on GPIO 32 and 33 and a simple quality index, such as the ratio of QRS energy to total energy, to reject windows that contain only noise [29].
5. **Tune the decision threshold for screening.** Lowering the threshold on the abnormal probability would raise the recall of SVEB and F beats at some cost in precision, which suits a screening tool.
6. **Move from beats to episodes.** Aggregating beat predictions over a minute, for example counting ectopic beats per minute, would give alerts that are closer to what a clinician wants to know and less sensitive to single errors.
7. **Revisit the thresholds and weights.** Base the bradycardia and tachycardia thresholds, the 0.7 / 0.3 risk weights and the cut-off of 60 on recorded scenarios rather than on judgement alone.
8. **Consider on-device inference.** A compact model, such as a small decision tree ensemble or a quantised neural network, could run on the ESP32 itself, which would keep basic screening working even when the hub is unavailable.

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GLOSSARY

Term	Definition
AAMI classes	The five heartbeat groups (N, SVEB, VEB, F, Q) recommended by the Association for the Advancement of Medical Instrumentation for reporting the performance of arrhythmia algorithms.
Balanced accuracy	The average of the recall of each class, so that small classes count as much as large ones.
Ectopic beat	A heartbeat that starts somewhere other than the sinus node, usually earlier than expected.
Edge computing	Processing data close to where it is produced, here on a local hub, instead of on a remote cloud server.
Fusion beat	A beat formed when a normal impulse and a ventricular impulse activate the ventricles at the same time.
Inter-patient evaluation	Testing a classifier on records from people whose beats were not used for training.
Macro F1	The unweighted average of the F1 scores of all classes.
Posture-aware fusion	Using the posture reported by the belt to decide whether a measured heart rate is explained by activity.
QRS complex	The sharp group of waves in the ECG produced by depolarisation of the ventricles.
R peak	The tallest point of the QRS complex, used to locate each beat.
RR interval	The time between two successive R peaks; its inverse gives the instantaneous heart rate.
Random Forest	An ensemble of decision trees, each trained on a bootstrap sample with random feature subsets, whose votes are combined.

APPENDICES

Appendix A: Test Scripts and Unit Test Output

The experiments in this thesis can be repeated with the scripts listed in Table A.1. All scripts read the dataset file from [34] or generate simulated ECG with NeuroKit2 [5], and use the random seed 42 where randomness is involved.

Table A.1: Scripts used for the experiments

Script	Purpose	Used in
arrhythmia_baseline_cv.py	Seven-classifier comparison with grouped five-fold cross-validation	Table 2.4
arrhythmia_training_replication.py	Deployed settings with and without class balancing	Table 2.5
exp_features.py	Feature extraction on simulated ECG, timing and edge cases	Section 3.1.5, Table 3.4
fix.py, wander.py	Current versus revised R-peak detector, noise and baseline wander	Tables 3.5 and 3.6
test_hub_logic.py	Unit tests of rules, fusion, screening output and risk score	Table 3.7

Source: Author.

The output of the unit test run is reproduced below.

Listing A.1: pytest output for test_hub_logic.py

```
test_hub_logic.py::test_brady_not_before_10 PASSED
test_hub_logic.py::test_brady_on_10th PASSED
test_hub_logic.py::test_brady_reset_by_normal PASSED
test_hub_logic.py::test_brady_per_device PASSED
test_hub_logic.py::test_zero_hr_ignored PASSED
test_hub_logic.py::test_tachy_boundary PASSED
test_hub_logic.py::test_missing_hr PASSED
test_hub_logic.py::test_fusion_running_suppressed PASSED
test_hub_logic.py::test_fusion_lying_not_suppressed PASSED
test_hub_logic.py::test_fusion_stale_posture PASSED
test_hub_logic.py::test_fusion_confidence_values PASSED
test_hub_logic.py::test_screening_veb PASSED
test_hub_logic.py::test_screening_normal PASSED
test_hub_logic.py::test_risk_examples PASSED
test_hub_logic.py::test_risk_emergency PASSED
test_hub_logic.py::test_risk_uses_max_engine PASSED
test_hub_logic.py::test_model_alone_needs_87 PASSED
============================== 17 passed in 0.06s
==============================
```

Appendix B: Feature Extraction and Revised R-Peak Detector

Listing B.1 gives the complete feature extraction function used on the hub, and Listing B.2 the revised R-peak detector proposed in Section 3.1.5.

Listing B.1: Beat feature extraction (backend/AI_models/arrhythmia/inference.py)

```
def _lead_features(signal, sampling_rate):
    if signal.size < int(sampling_rate * 2.5) or np.ptp(signal) < 1e-6:
        return None
    normalized = (signal - np.median(signal)) / (np.std(signal) + 1e-8)
    distance = max(1, int(0.25 * sampling_rate))
    peaks, _ = find_peaks(normalized, distance=distance, prominence=0.45)
    if len(peaks) < 3:
        return None
    r = int(peaks[-2])
    pre_rr = float(peaks[-2] - peaks[-3])
    post_rr = float(peaks[-1] - peaks[-2])
    radius = max(2, int(0.18 * sampling_rate))
    left, right = max(0, r - radius), min(len(normalized), r + radius + 1)
    beat = normalized[left:right]
    q_rel = int(np.argmin(beat[: max(1, r - left + 1)]))
    s_start = min(len(beat) - 1, r - left)
    s_rel = s_start + int(np.argmin(beat[s_start:]))
    p_area = normalized[max(0, r - int(.25 * sampling_rate)):
                        max(1, r - int(.08 * sampling_rate))]
    t_area = normalized[min(len(normalized), r + int(.08 * sampling_rate)):
                        min(len(normalized), r + int(.40 * sampling_rate))]
    p_peak = float(np.max(p_area)) if p_area.size else 0.0
    t_peak = float(np.max(t_area)) if t_area.size else 0.0
    q_index, s_index = left + q_rel, left + s_rel
    positions = np.linspace(left, right - 1, 5).astype(int)
    return {
        "pre-RR": pre_rr, "post-RR": post_rr,
        "pPeak": p_peak, "tPeak": t_peak, "rPeak": float(normalized[r]),
        "sPeak": float(normalized[s_index]), "qPeak":
float(normalized[q_index]),
        "qrs_interval": float(max(1, s_index - q_index)),
        "pq_interval": float(max(1, q_index - (r - int(.18 *
sampling_rate)))),
        "qt_interval": float(max(1, (r + int(.36 * sampling_rate)) -
q_index)),
        "st_interval": float(max(1, int(.08 * sampling_rate))),
        **{f"qrs_morph{i}": float(normalized[p]) for i, p in
enumerate(positions)},
    }
```

Listing B.2: Revised R-peak detector

```
from scipy.signal import butter, filtfilt, find_peaks
B, A = butter(2, [5 / 125, 15 / 125], btype="band")      # 5-15 Hz at 250 Hz

def detect_r_peaks(normalized, fs=250):
    energy = filtfilt(B, A, normalized) ** 2
    peaks, _ = find_peaks(energy, distance=int(0.25 * fs),
                          height=0.3 * np.percentile(energy, 99))
    edge = int(0.15 * fs)                                  # drop cut-off
beats
    return peaks[(peaks >= edge) & (peaks < len(normalized) - edge)]
```

Appendix C: Work Breakdown Structure and Gantt Chart

The work on the component was organised into four work packages: (1) requirements, literature review and design; (2) dataset analysis, feature extraction and model training; (3) hub rules, posture fusion and integration with the group's modules; and (4) testing, evaluation and thesis writing. Figure C.1 shows the schedule.

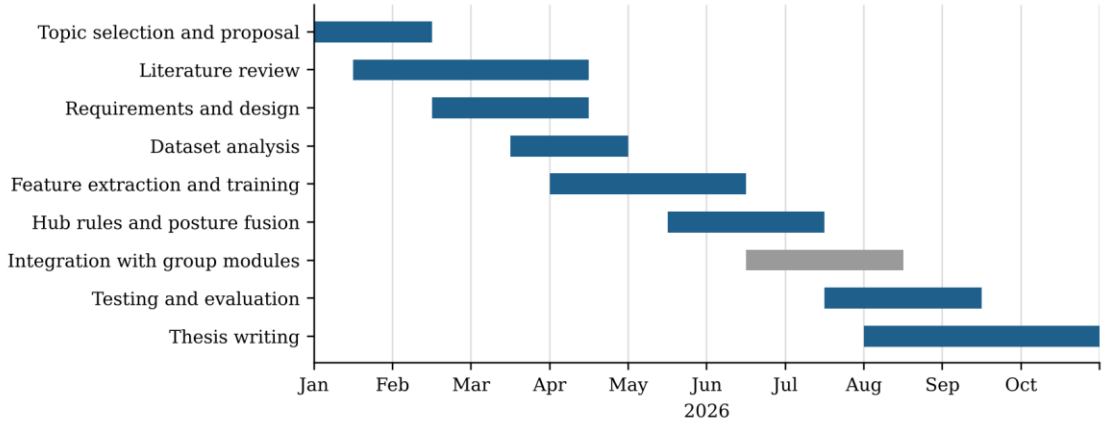


Figure C.1: Gantt chart of the arrhythmia component work

Appendix D: Academic Integrity and AI Use Disclosure

Generative AI was used in preparing this thesis as described in Table D.1, in line with the module policy.

Table D.1: Use of generative AI tools

Tool	Purpose	Verification by the author	Sections affected
Claude (Anthropic)	Structuring the thesis to the SLIIT guidelines; drafting and editing text from the author's code, results and project documents; running the comparison and simulation scripts; drawing figures	The author supplied the code, data and results, checked every number against the source output, reviewed and rewrote the text, and can reproduce the experiments with the scripts in Appendix A	All chapters, Appendices A to C

Source: Author.

The author remains responsible for the content of this thesis, including every number, claim and reference.